Chapter 2 · Solutions

P04 · Trace the expression

Hint. Match Q first. Do not discard duplicates before projecting.

The natural join has five rows:

P	Q	R
1	a	9
1	a	10
2	a	9
2	a	10
3	b	10

R=10 keeps three rows. Their projected Q values are a,a,b, which form the set {a,b}. The final cardinality is 2. After removing (b,10), the first four joined rows remain. Selection leaves (1,a,10),(2,a,10), and projection returns {a}, with cardinality 1. The product sizes are 3×3=9 and 3×2=6.

Check. A join matches rows. A product makes every pair. Set projection removes duplicate projected tuples.

P05 · Count matching pairs

Hint. Group each side by the joining value.

Value 1 contributes 2×3=6 pairs. Value 2 contributes 1×1=1. Value 3 has no match. Total: 7. After making the S join column unique, value 1 contributes 2 and value 2 contributes 1, giving 3.

The bound |R| requires uniqueness on S's joining value. It does not apply to the original S, where each value-1 row of R has three partners.

P06 · Every required course

Hint. Make one set of courses per candidate student.

Student 1 has {M,P}; student 2 has {M}. The nonempty requirement is met only by 1. When Required is empty, both candidates from Enrolment pass, so classical division returns {1,2}.

A query starting with Student has candidate domain {1,2,3}. No candidate can miss a member of an empty Required set. It returns {1,2,3}, including student 3. The outputs differ because their candidate domains differ.

P07 · Disprove the claims

Hint. One name and an empty relation are enough.

For (a), choose Team={Asha}, Defender={Asha}, Forward=empty. The union covers Asha, so the first difference is empty. The intersection is empty, so the second difference contains Asha.

For (b), choose Team={Asha} and both player sets empty. Their union minus Team is empty. But Team minus their union contains Asha, so Team coverage fails.

Check. T minus U empty means T⊆U. U minus T empty means U⊆T. They are different directions.

P08 · Output rows versus witnesses

Hint. Build the witness list, then count distinct qualifying output tuples.

For >1, row (1,a) has witnesses (1,10),(1,20), and (2,b) has (2,30). Thus there are 3 witness pairs but 2 output tuples. Row (3,c) has no qualifying witness.

For ≥1, (3,1) also becomes a witness. There are now 4 witness pairs and 3 output tuples. The additional z condition is always satisfied by z=x. It changes neither result.

Check. Distinct variable names do not impose distinct values. An explicit inequality would be needed.